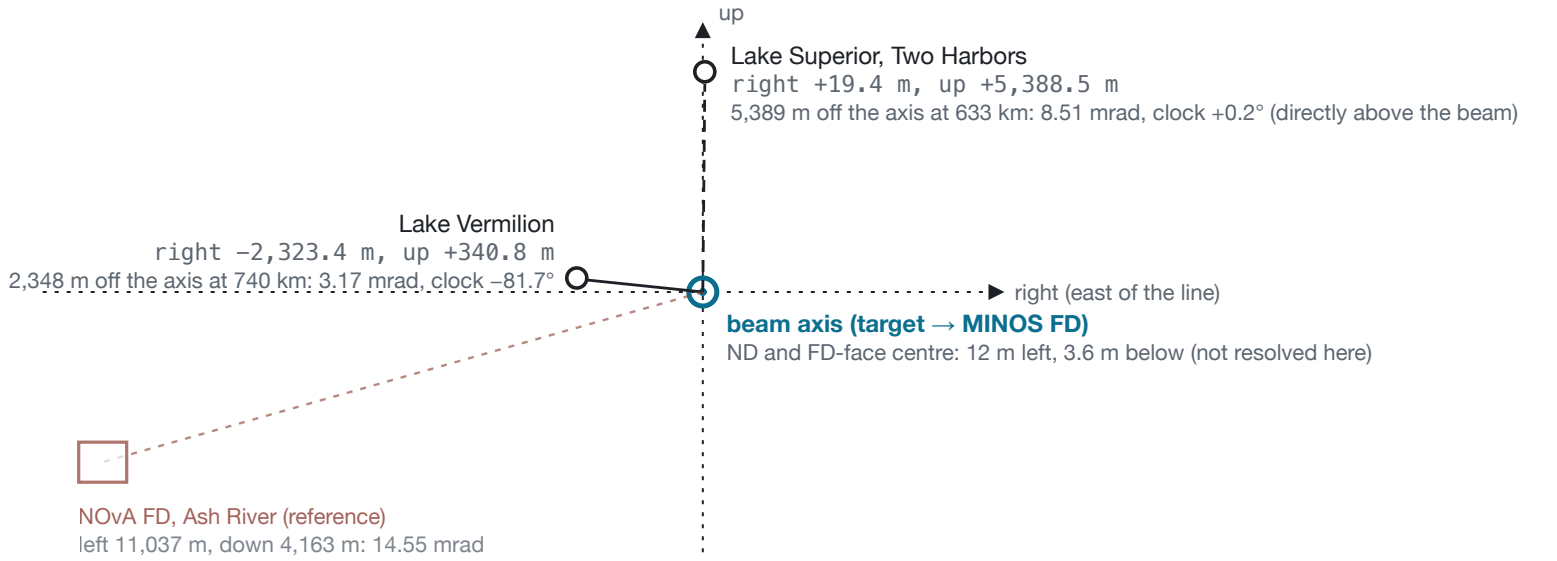
up
Lake Superior, Two Harbors
right +19.4 m, up +5,388.5 m
5,389 m off the axis at 633 km: 8.51 mrad, clock +0.2° (directly above the beam)
Lake Vermilion
right −2,323.4 m, up +340.8 m
2,348 m off the axis at 740 km: 3.17 mrad, clock −81.7°
right (east of the line)
beam axis (target → MINOS FD)
ND and FD-face centre: 12 m left, 3.6 m below (not resolved here)
NOvA FD, Ash River (reference)
left 11,037 m, down 4,163 m: 14.55 mrad
transverse offsets drawn to a common scale (1 km ≈ 20 px)